

Mathematical Formulation of NDI and Robot Pose Optimization

1. Problem Definition

Given the known target lesion position and puncture path, this system aims to find the optimal NDI tracker pose and robot base position to maximize marker visibility while satisfying spatial geometric constraints. This is a **Non-linear Constrained Optimization Problem**.

Known Input: Pose of the target puncture trajectory point $T_{target} \in SE(3)$

Variables to Solve:

- NDI camera pose: $X_{ndi} \in SE(3)$
- Robot base pose: $X_{robot} \in SE(3)$

2. Objective Function

Due to the inclusion of 3D mesh occlusion and Volume checks, this system cannot be expressed by a simple continuous analytical function $f(x)$. Instead, it minimizes a system cost function J that includes an Occlusion Penalty and a Solid Angle:

$$(X_{ndi}^*, X_{robot}^*) = \arg \min_{X_{ndi}, X_{robot}} J(X_{ndi}, X_{robot} \mid T_{target}) \quad (1)$$

3. Constraints

The above optimization must strictly satisfy the following geometric and kinematic conditions:

$$\textbf{Reachable:} \quad IK(X_{robot}, T_{target}) \neq \emptyset, \quad q_{min} \leq q_i \leq q_{max} \quad (2)$$

$$\textbf{Collision-free:} \quad \text{Collision}(X_{robot}, q) = \text{False} \quad (3)$$

$$\textbf{Visibility:} \quad M_i(q) \in \text{Volume}(X_{ndi}), \quad \text{Raycast}(X_{ndi}, M_i(q)) = \text{Clear} \quad (4)$$

where IK is the inverse kinematics solution, q represents the joint angles for each dimension, and $M_i(q)$ represents the spatial coordinates of the NDI markers at that pose.

4. Numerical Solution Method

Since the Volume and Raycast checks are discontinuous Boolean operations (non-differentiable), the method currently used by the program is **Grid Search / Sampling**:

$$J_{min} \approx \min_{(X_{ndi}, X_{robot}) \in S} J(X_{ndi}, X_{robot}) \quad (5)$$

In this equation, S represents the set of all possible installation space samples (i.e., `robot_x_list` and `ndi_x_positions` in the script).

Let $\mathcal{N}$ be the set of NDI camera positions and $\mathcal{R}$ be the set of robot base positions, with sizes $|\mathcal{N}| = n$ and $|\mathcal{R}| = m$. We evaluate all $n \times m$ spatial configurations.

To optimize simulation throughput, the combinations are processed as m sequential batches. In each batch, a fixed robot base position $r \in \mathcal{R}$ is used, and n parallel environments are spawned simultaneously to evaluate all n NDI positions in $\mathcal{N}$.